

COMP9417 Group Project

Group Six Seven

Introduction

Air pollution is an increasingly important public health concern, necessitating the creation of reliable methods of forecasting air quality indicators. This report evaluates the effectiveness of a selection of regression- and classification-based machine learning models in predicting future air pollution, in conjunction with informed feature representations to test model robustness. We use the Air Quality Dataset from the UCI Machine Learning Repository [3], collected via an urban roadside multi-sensor for the air concentration of five metal oxides over 2004-2005.

Our models take eight hourly time series (the five sensor streams, temperature, relative humidity, and absolute humidity) over hours $1, \dots, t$, in order to predict the concentration of five pollutants of interest (see appendix A) at hours $t + 1, t + 6, t + 12, t + 24$.

Data Analysis

Data Cleaning

Empty columns and rows were removed, dots in timestamps were converted to colons, commas in floats were converted into dots, and entries of -200 were converted into NaNs.

Variable Relationships and Patterns

We produced scatter plots and computed correlations between variables of interest. The points were divided into 5 temporal windows and colour-coded to also investigate the presence of sensor drift. The full set of scatters and correlations can be found in the appendix.

While many variable pairs display strong linear correlation (subplot 1, fig 1), some other variable pairs have an observable relationship that is not linear (subplot 2, fig 1). This later motivates data imputation via kNN instead of linear regression. We also observed a light degree of sensor drift in some variable pairs (subplot 3, fig 1). Lastly, while temperature and relative humidity were weakly related to the pollutant levels, absolute humidity did seem to affect some variables, such as the NO_2 sensor data appearing to be capped by a linear function of absolute humidity (subplot 4, fig 1).

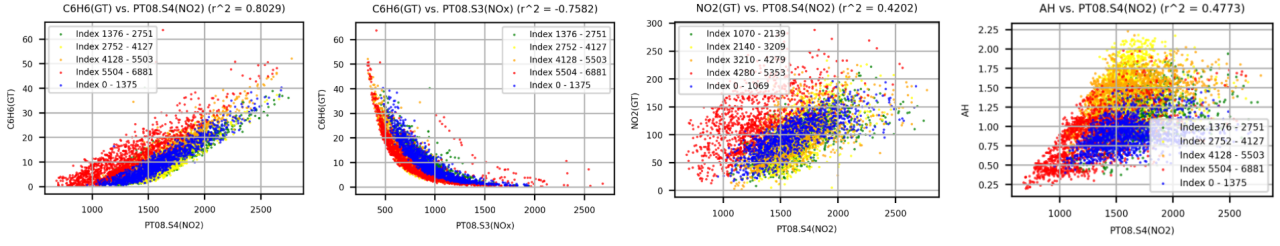
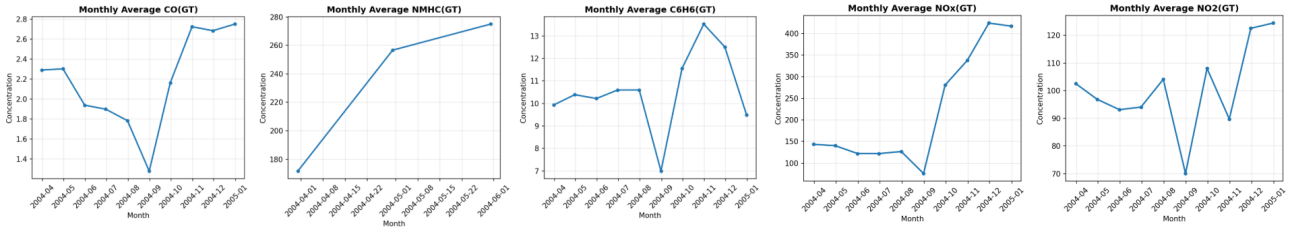


Figure 1: A selection of scatter plots of variable pairs

Data Quality and Missing Values

We first tallied up how often missing-value gaps of different lengths occur (see appendix B). We then grouped variables based on shared missingness behaviour and analysed windows of 5 or more missing entries (see appendix B). Our main findings are: **1)** Most data gaps consist of just 1 missing data point. There are 814 1-gaps, while longer gaps have almost all had single-digit occurrences. **2)** The data for NMHC(GT) is missing past May 2004, displaying a gap of 8126 hours (around 339 days). That is, we have less than two months of workable data. **3)** The streams for the five sensors, T, AH, RH, and C₆H₆(GT) go missing at the same times. Also, NO_x(GT) and NO₂(GT) appear to share very similar missing data patterns.

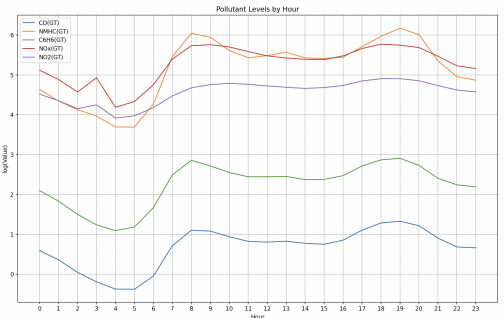
Seasonal Effects



The monthly average concentrations of each pollutant were plotted. Apart from NMHC(GT), whose incomplete seasonal pattern is due to missing data, all the other pollutants appear to follow a seasonal trend, decreasing in concentration in the warmer months and increasing in the colder months, which is likely due to an increase in combustion activity.

Daily Time Patterns

We took the average concentration of each pollutant at each hour across all days in 2004, and organised this data in a log-transformed plot. We observe that all pollutants demonstrate elevated concentrations during daylight hours, with slight peaks at 8-9 am and 7-8 pm. This can be explained by increased traffic emissions during daytime, especially during peak hours. On the other hand, the daily minimums typically occur during hours of minimal human activity, at around 4-5 am.



Methodology

Preprocessing

Features were derived for hour, weekday, month, timestamp. For missing values, we imputed using a mixture of kNN, cubic spline, and 30-day windowed hourly average, based on a ranking of expected RMSEs from testing. The testing involved imputing artificially removed gaps of sizes 1 to 9 in an otherwise intact data window of 17. We then deployed imputation methods which displayed an s.d. RMSE of less than 0.5, with the 30-day windowed hourly average as a fallback. kNN was chosen due to the nonlinear but strong relationships found earlier. The five sensor streams were max-min normalised. We chose this instead of z-scores since the presence of sensor drift may have meant that a z-score of 0 may not be truly representative of the mean at particular times.

Feature Engineering

Because pollutants often linger in the air for an extended period after being released into the atmosphere, we considered the moving average of the pollutants with windows of 6, 12, and 24 hours. Those window sizes were specifically chosen to predict future pollutant concentrations for 1, 6, 12 and 24-hour horizons.

Furthermore, for each measured variable and moving average, we also implemented lagged variables up to 3 hours beforehand, again adhering to the idea that pollutants linger in the air for extended periods of time.

Anomaly Detection in Dataset

Using the classical definition for outliers, we identified anomalies by taking values that were more than 1.5 times the IQR away from Q1 and Q3. For all pollutants except NO₂, outliers often occurred consecutively, and 80% of outliers occurred during the morning or evening. This is likely due to vehicle emissions during commuting hours, since this data was taken at the road level. NO₂ levels are the exception, with 89% of outliers during sleeping hours. These outliers are believed to be caused by the delayed breakdown of NO_x-related pollutants into NO₂ after they were emitted during the evening commute. Additionally, unusually high concentrations of pollutants occur mainly during spikes in relative humidity, which occur during lower temperatures.

Precision-Recall Tradeoffs

Precision is defined as $\frac{\text{true positives}}{\text{true positives} + \text{true negatives}}$ and recall is defined as $\frac{\text{true positives}}{\text{true positives} + \text{false negatives}}$. This gives rise to the precision recall trade-off. This is because, for instance, improving precision means reducing false positives, which incentivise over-predicting the negative class and thus worsens recall. This is similarly true in the opposite direction. However, this trade-off is not

strict, as a model may simply be made more accurate, improving both precision and recall. As such, anomaly detection enhances model accuracy, reliability and model robustness since it reduces the impact of one-off events like equipment failure or rare meteorological events, and allows the model to account for previously seen anomalies.

For air pollution, overcautious predictions are preferable to under-predicting to prevent a 'false sense of security'. Thus, improving recall is favoured over precision.

Model Selection Rationale

Firstly, the naive baseline for the sake of comparison was chosen to be one that used the ground truth concentration at t for the prediction at $t+1$, $t+6$, $t+12$, and $t+24$. Note that this is not realistic as a deployed model since it requires access to the ground truth reference analyser streams, which are not available to the deployed model.

The rationale for classification and regression models is identical since classification reduces to determining which 'bin' a continuous value falls into. A moderate dataset size gives the freedom to use computationally expensive models.

Linear regression models were chosen for their simplicity and interpretability, despite not being able to capture nonlinear behaviour. Ordinary Least Squares (OLS) regression is used as a baseline with Lasso regression for feature selection to reduce nonzero coefficients.

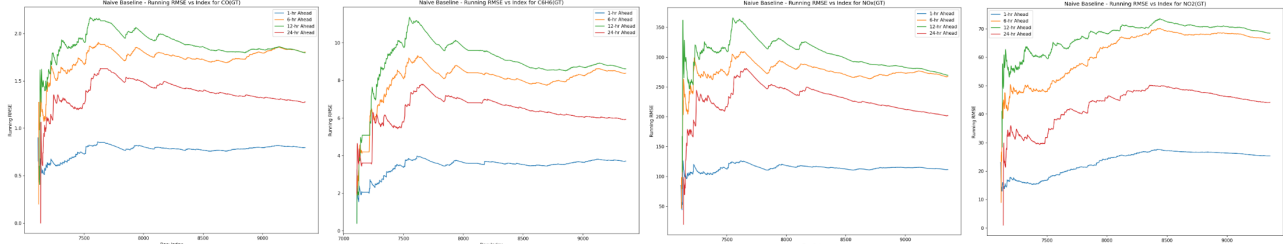
Tree-based ensemble methods, namely Random Forest and Gradient Boosting, were selected for their feature ranking ability. Random Forest aggregates trees to improve generalisation, reducing overfitting. Bootstrapping was used to reduce variance and increase robustness to sensor noise. Gradient Boosting reduces bias by minimising residuals, making it a good choice for this project. However, its overfitting tendency makes accounting for noise with specialised hyperparameters a consideration.

Finally, Long Short-Term Memory (LSTM) neural networks were chosen as the data is in a time series. LSTMs learn delayed effects without manually lagging variables well. This will handle non-linear temporal patterns better than other models. A single LSTM model with 4 outputs corresponding to $t + 1, t + 6, t + 12, t + 24$ is instantiated.

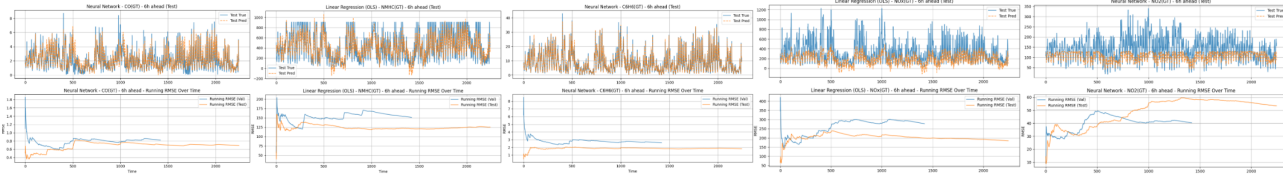
Results - RMSE's of Regression Models

	Model	CO_1hr	NMHC_1hr	C6H6_1hr	NOx_1hr	NO2_1hr	CO_6hr	NMHC_6hr	C6H6_6hr	NOx_6hr	NO2_6hr	CO_12hr	NMHC_12hr	C6H6_12hr	NOx_12hr	NO2_12hr	CO_24hr	NMHC_24hr	C6H6_24hr	NOx_24hr	NO2_24hr
0	Gradient Boosting	0.58	27.38	0.20	160.38	53.35	0.87	143.64	3.46	195.96	56.54	1.03	203.35	4.49	211.42	61.19	1.22	232.06	5.57	219.65	64.52
1	LSTM-1	0.81	214.51	3.39	200.75	68.49	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan
2	LSTM-12	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	1.12	255.70	5.35	239.92	68.84	nan	nan	nan	nan	nan
3	LSTM-24	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	1.15	248.99	5.42	225.36	68.65
4	LSTM-6	nan	nan	nan	nan	nan	1.10	246.68	4.90	227.55	68.72	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan
5	Lasso Regression	0.64	109.52	1.36	175.88	53.63	0.90	125.62	2.62	184.95	55.07	1.12	166.54	4.07	200.91	58.87	1.25	205.41	5.58	215.97	63.91
6	Linear Regression (OLS)	0.61	109.23	1.37	175.76	53.58	0.74	125.61	2.50	184.29	54.54	0.87	165.19	3.62	199.74	58.05	1.11	204.79	5.10	215.03	63.40
7	Neural Network	0.59	241.10	0.18	198.82	53.20	0.69	249.54	1.80	209.51	53.50	0.94	259.93	3.18	224.01	55.62	1.22	270.49	5.09	229.15	68.48
8	Random Forest	0.59	34.95	0.16	162.02	53.12	0.77	145.34	2.45	189.30	56.59	1.03	192.73	3.55	202.47	59.18	1.28	233.11	5.46	221.44	63.93
9	Naive	0.80	nan	3.71	111.78	25.37	1.80	nan	8.39	267.59	66.51	1.80	nan	8.62	269.66	68.53	1.28	nan	5.94	201.94	44.24

The evolution of RMSE's for the naive model are as follows:



Meanwhile, our best models for the 6 hour horizons demonstrated these evolutions of RMSE's:



Accuracy of Classification Models

	Model	CO_low_1hr	CO_mid_1hr	CO_high_1hr	CO_low_6hr	CO_mid_6hr	CO_high_6hr	CO_low_12hr	CO_mid_12hr	CO_high_12hr	CO_low_24hr	CO_mid_24hr	CO_high_24hr
0	NeuralNetwork	0.91	0.90	0.88	0.87	0.87	0.78	0.73	0.81	0.71	0.66	0.71	0.60
1	RandomForest	0.88	0.93	0.87	0.82	0.90	0.75	0.68	0.74	0.70	0.64	0.75	0.64
2	Naive	0.86	0.76	0.86	0.56	0.57	0.60	0.54	0.60	0.60	0.76	0.66	0.76

Discussion

Comparison of methods and analysis of feature importance

The 1 hour prediction regime is dominated by the Naive model and Gradient Boosting. The Naive model's performance here is expected since pollutant levels change little over 1hr, and it has access to ground truths that other models are not given, so it is not discouraging in terms of other model performance. Gradient boosting worked well, sometimes outperforming the Naive model. This is likely due to it's ability to reduce variance by aggregating the outputs of weaker models which were able to latch onto the relatively small changes in the 1hr regime compared to other models.

The 6hr and 12hr regimes were predicted best by Linear Regression and a Neural Network. Linear Regression is a simpler model, suggesting a linear relationship between previous pollutant levels and future levels for some pollutants. Neural networks are more complex, capturing nonlinear relationships. This model was too complex and tended to visually overfit some data, but was useful for pollutants that exhibited a more complex relationship such as CO, C6H6 and NO2.

The 24hr regime is still dominated by Linear Regression and Neural network. However, the Naive model also worked well. Similar to the 1hr case, the Naive model likely performs well due to daily cycles in pollutant behaviour from human activity. Again, this is due to the Naive model having access to ground truths which other models do not.

Finally, The multi-output LSTM also underperformed. Although LSTMs are well-suited for sequential data, this dataset is relatively small, noisy, and affected by sensor drift—conditions under which recurrent models struggle to learn stable temporal patterns. Random Forest models performed reasonably across all regimes despite not being a leader in any category. The ensemble methodology makes it able to capture most behaviour with a high reduction in variance without focusing strongly on bias, as the highly erratic nature of the data implies that we must focus on generalising, as a low bias will likely result in overfitting.

Metrics and Appropriateness

Regression: Other metrics for regressors have their own pros and cons. Mean absolute error is less sensitive to outliers than RMSE due to absence of the square term, but it also penalises outliers less. Thus, RMSE served as a well-rounded and intuitive metric for our purposes.

Classification: Metrics other than accuracy could have been appropriate. In a dataset with few positive instances, a model that always predicts the negative class will have high accuracy. Here, recall could be a better metric if these forecasts are for public health warnings, where failing to predict periods of high pollutant concentration can be detrimental, since recall measures how many positive instances are correctly predicted.

Limitations and Future Improvements

Sensor Drift: As seen in the exploratory scatter plots, this dataset has signs of sensor drift. To improve model robustness for prolonged deployment, future work should investigate sensor drift detection techniques, including exponentially weighted moving average (EWMA) and cumulative sum (CUSUM) [1], and adaptive learners such as online stochastic gradient descent.

Temporal concept drifts: To capture temporal drifts, future work could add a feature for the fraction of each day that receives sunlight, to capture seasonal variations in sunlight duration and its meteorological effects, so deployed models remain accurate across seasons. Furthermore, due to ongoing global warming [2], an extended dataset could allow prediction of multi-year temporal drifts in pollutant patterns.

Conclusion

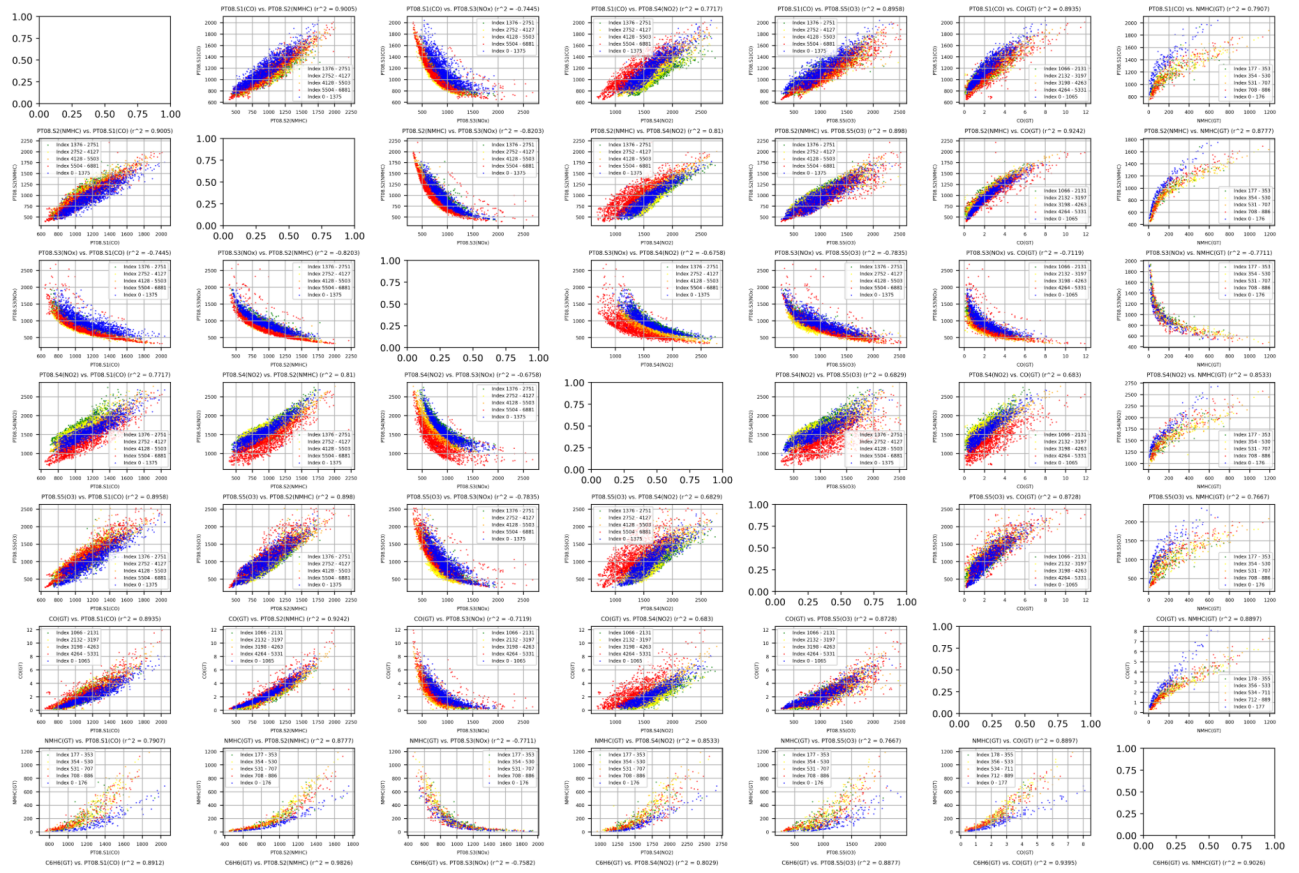
This project assessed several machine learning models for predicting air pollutant concentrations using the provided UCI Air Quality data set. Despite shortcomings in the data (specifically significant missingness in the NMHC(GT) variable), our methodology, which combined imputation, normalisation and feature engineering, enabled the models to accurately capture short-term pollutant dynamics. We see that for the Linear Regression and Neural Network models tend to have the lowest RMSE for the regression models. Meanwhile, for classification models, it appears that neural networks again provide the most accurate results.

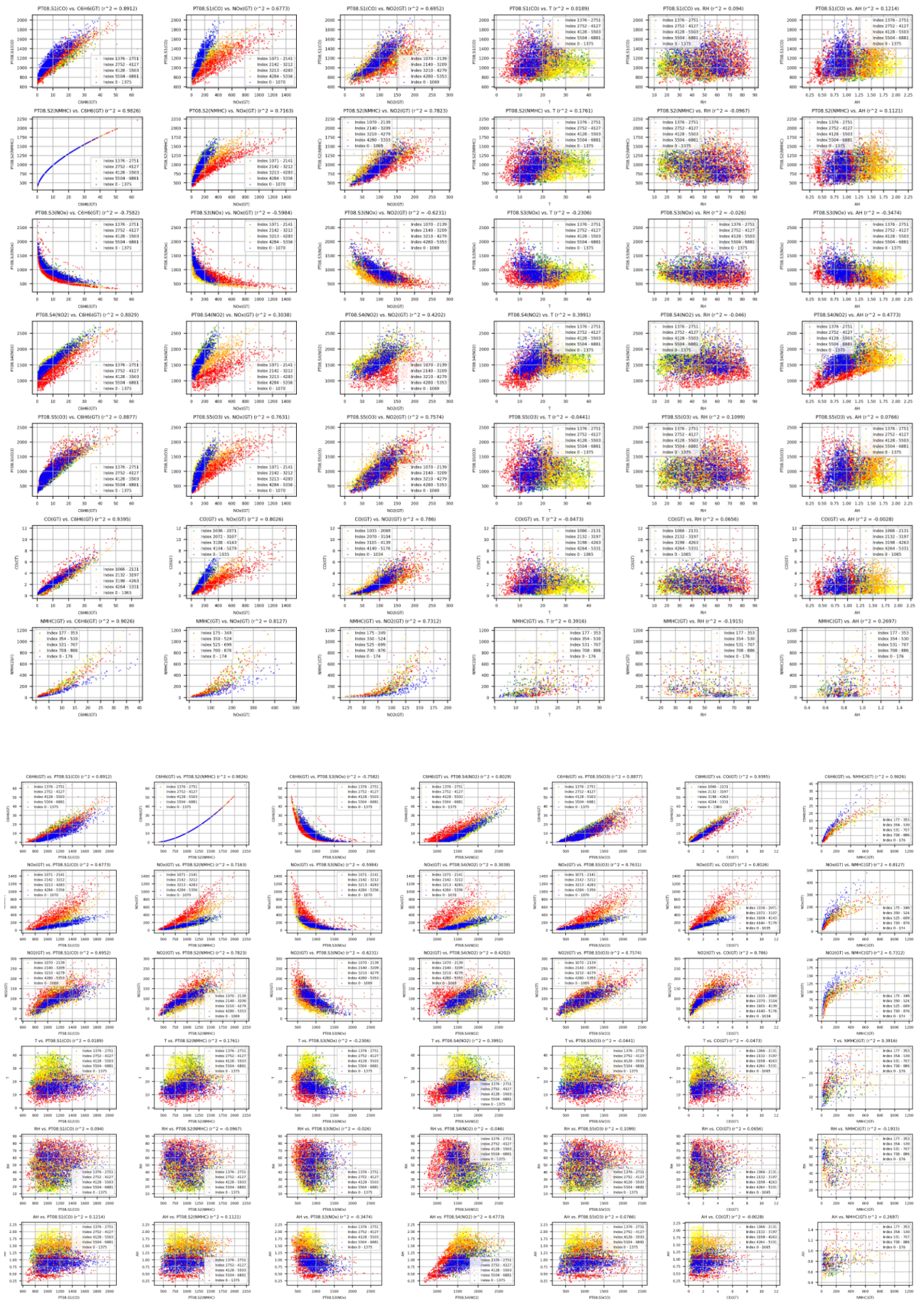
Appendices

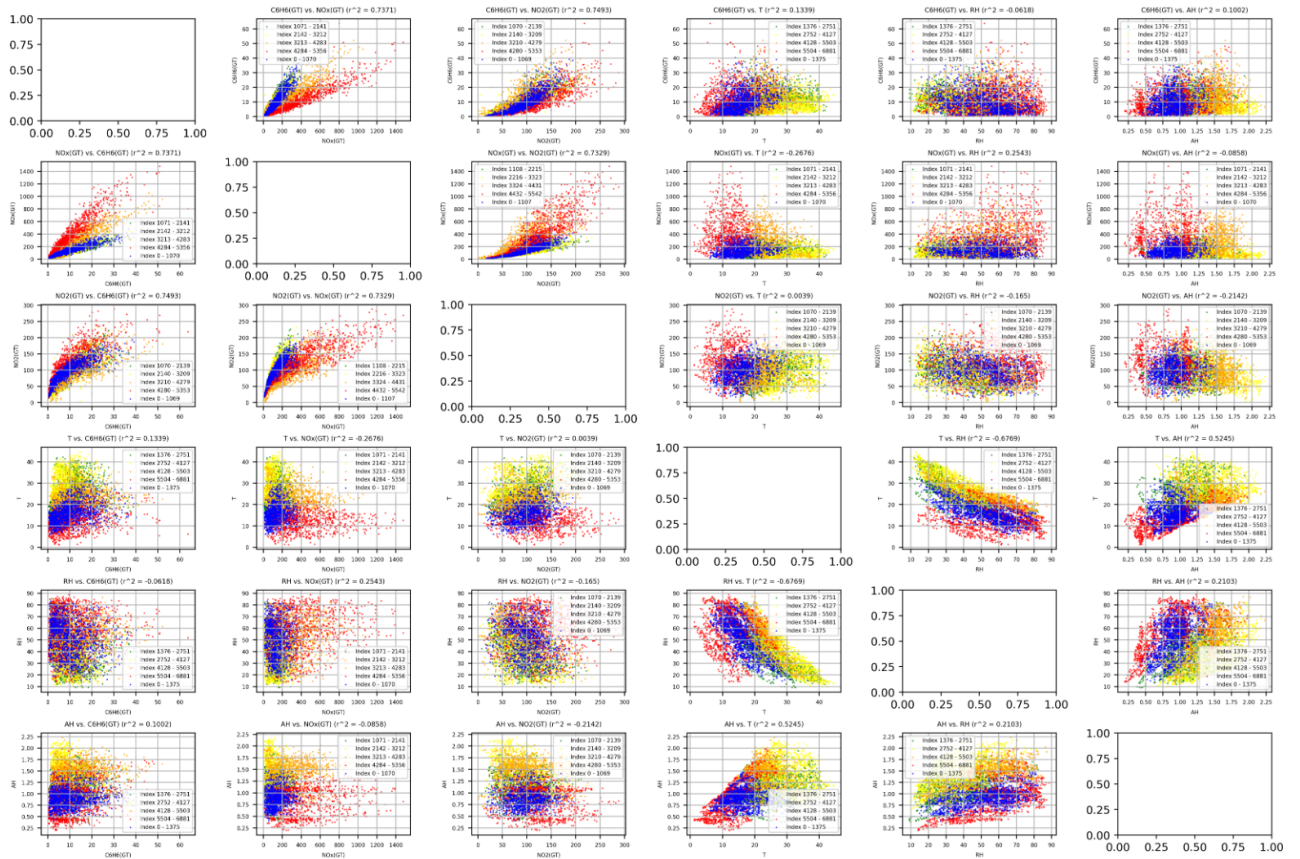
Appendix A - Dataset Variables Information

Variable Name	Role	Type	Description	Units
Date	Feature	Date		
Time	Feature	Categorical		
CO(GT)	Feature	Integer	True hourly averaged concentration CO in mg/m ³ (reference analyzer)	mg/m ³
PT08.S1(CO)	Feature	Categorical	hourly averaged sensor response (nominally CO targeted)	
NMHC(GT)	Feature	Integer	True hourly averaged overall Non Metanic HydroCarbons concentration in microg/m ³ (reference analyzer)	microg/m ³
C6H6(GT)	Feature	Continuous	True hourly averaged Benzene concentration in microg/m ³ (reference analyzer)	microg/m ³
PT08.S2(NMHC)	Feature	Categorical	hourly averaged sensor response (nominally NMHC targeted)	
NOx(GT)	Feature	Integer	True hourly averaged NOx concentration in ppb (reference analyzer)	ppb
PT08.S3(NOx)	Feature	Categorical	hourly averaged sensor response (nominally NOx targeted)	
NO2(GT)	Feature	Integer	True hourly averaged NO2 concentration in microg/m ³ (reference analyzer)	microg/m ³
PT08.S4(NO2)	Feature	Categorical	hourly averaged sensor response (nominally NO2 targeted)	
PT08.S5(O3)	Feature	Categorical	hourly averaged sensor response (nominally O3 targeted)	
T	Feature	Continuous	Temperature	°C
RH	Feature	Continuous	Relative Humidity	%
AH	Feature	Continuous	Absolute Humidity	

Appendix A - Scatter plots for all variable pairs







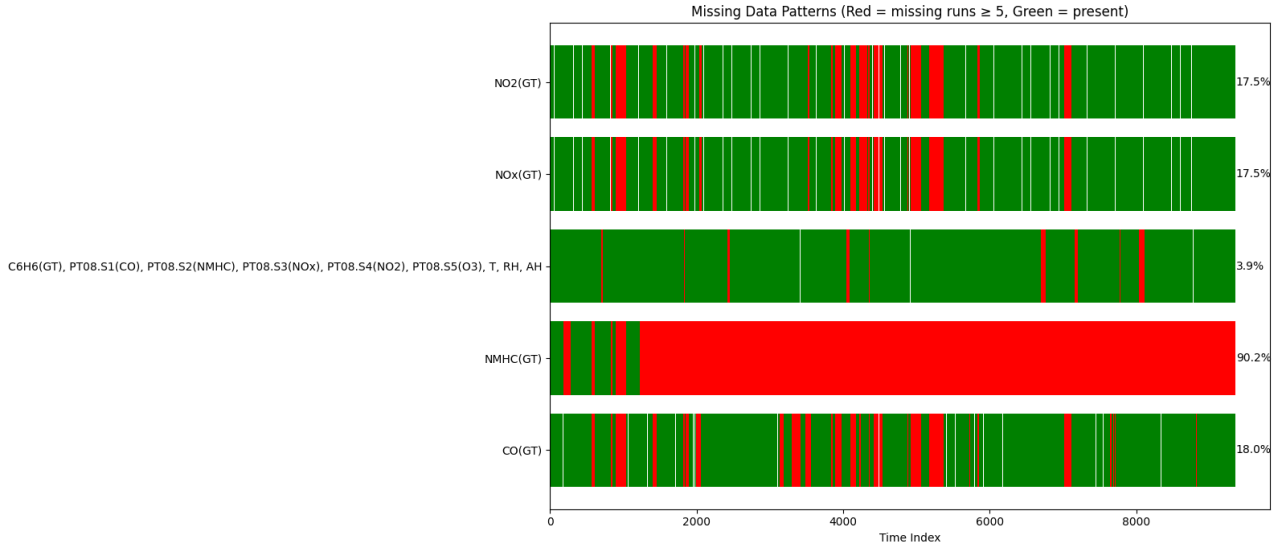
Appendix B - Frequency table of data gaps of different lengths

==== FIG 1: GAP LENGTH FREQUENCY TABLE ====

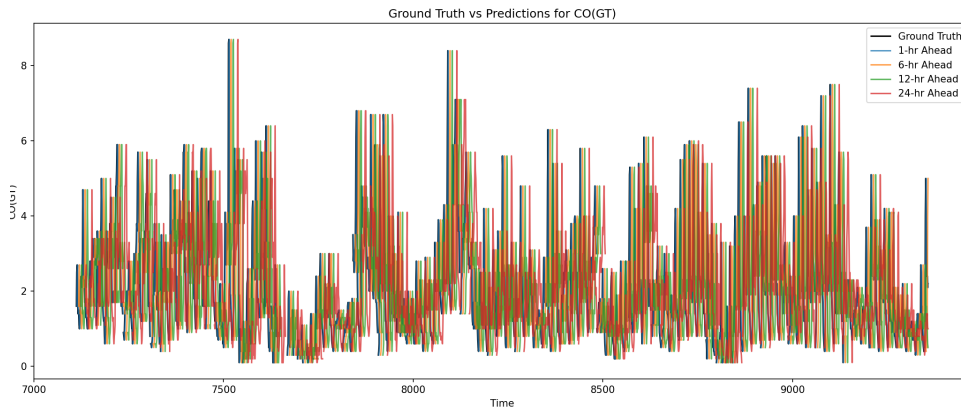
1	814
2	14
3	10
4	13
5	9
6	1
8	10
9	10
10	10
11	1
12	2
14	9
18	1
19	4
20	4
24	12
25	1
29	3
30	2
31	1
32	2
33	2
38	9
39	3
41	2
45	9
46	3
48	4
52	9
54	4
57	3
63	1
72	1
73	2
75	9
76	10
84	2
86	1
96	3
102	1
116	2
122	1
142	3
146	4
173	3
8126	1

Name: count, dtype: int64

Appendix B - Missing data windows, showing critical issue in NMHC(GT) and shared missingness



Appendix C - Naive regression model, predicted vs observed values



References

- [1] Alireza Estaji, Maximilian Götzinger, Benedikt Tutzer, Stefan Kollmann, Thilo Sauter, and Axel Jantsch. Evaluation of drift detection algorithms in the condition monitoring domain. *IEEE Transactions on Industrial Informatics*, PP:1–10, 01 2024.
- [2] Rebecca Lindsey and LuAnn Dahlman. Climate change: Global temperature, 2025. Accessed 6 July 2025. Available at: <https://www.climate.gov/news-features/understanding-climate/climate-change-global-temperature>.
- [3] Saverio Vito. Air Quality. UCI Machine Learning Repository, 2008. DOI: <https://doi.org/10.24432/C59K5F>.

Other Resources Used

Background theory on machine learning models and evaluation metrics was drawn from UNSW COMP9417 25T3 lecture slides.

Our codebase extensively uses Python machine learning and data processing libraries, including `pandas`, `numpy`, `matplotlib`, `scikit-learn`, `tensorflow`, `keras`.